Privacy-Preserving Data Analytics with Generative AI

A Metadata-Driven Secure Prompting Approach

Group Activity

Time: 45 minutes

All groups will use the following synthetic mental-health dataset from SynthDataHub:

GitHub file page:

https://raw.githubusercontent.com/ntious/SynthDataHub/main/Healthcare/Mental%20Health/synthetic_mental_health_dataset.csv

Download the CSV and load it into your local Python, Jupyter, GitHub Codespaces, or Google Colab environment.

Do not upload the CSV file directly to a GenAI system.

Although the dataset is synthetic, treat it as though it contains real healthcare information. The dataset includes direct identifiers, demographic attributes, mental-health information, treatment information, free text, substance-use information, genetic-marker information, and an outcome variable.

Your group must identify which metadata can be shared with GenAI and which metadata should be:

- excluded;
- aliased;
- generalized;
- suppressed;
- or retained with caution.

At minimum:

- exclude Patient_ID;
- exclude Life_Events;
- do not share raw values from any column;
- do not provide category-frequency tables containing small groups;
- and carefully review health, genetic, demographic, treatment, and substance-use fields before including them in sanitized metadata.

Analytical Questions

Use the sanitized metadata to ask a GenAI tool to generate local Python code addressing these questions:

1. Which approved variables are most strongly associated with Outcome_Variable?
2. Can Outcome_Variable be predicted using approved features without using Patient_ID or Life_Events?

The generated notebook must include:

- local schema validation;
- aggregate descriptive statistics;
- at least one privacy-safe visualization;
- one appropriate analytical or predictive model;
- aggregate evaluation results;
- and a short interpretation.

Do not require an advanced or highly optimized model. A correctly validated basic model is sufficient for this activity.